



Privacy in the Rearview: Empirical Analysis of Connected Vehicle Data Exposure

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RESEARCH QUESTIONS

- RQ1:** What data can we observe connected vehicles and their companion mobile apps sharing about their users, particularly personally identifiable information (PII) and sensitive data linked to PII (e.g. location)?
- RQ2:** Who receives connected vehicle data, whether directly from the vehicle or from companion smartphone apps, particularly when it comes to third parties?
- RQ3:** Where is this data transmitted, including international destinations beyond the US, given that all of our test vehicles were US models operated domestically?

DATA + MEASUREMENT

App Tests

Done via physical iPhones running a variety of iOS versions and a laptop running mitmproxy. These companion apps were linked to a physical car. Some apps performed certificate pinning on domains that were critical for user account authentication. We circumvented this restriction using mitmproxy's "ignore hosts" option to avoid intercepting these domains. For the apps, we were able to see the actual data being sent for the domains we were able to intercept.

Wi-Fi Tests

	Baseline	Active	Driving
Test	Connect to Wi-Fi, leave vehicle alone 30 minute duration	Interact with all vehicle functions Variable duration (>15 mins)	Drive private service road. At least one instance of hard braking. Max speed of ~40MPH 15 minute duration
Purpose	How much does the vehicle send when it is "off"? With whom does it communicate?	How does the vehicle communicate with regular use?	Does the vehicle get more chatty when we drive? Does it contact new domains?

Limitation: We cannot see the actual data being sent: Only the domains contacted and the size+rate of sending.

Figure 1

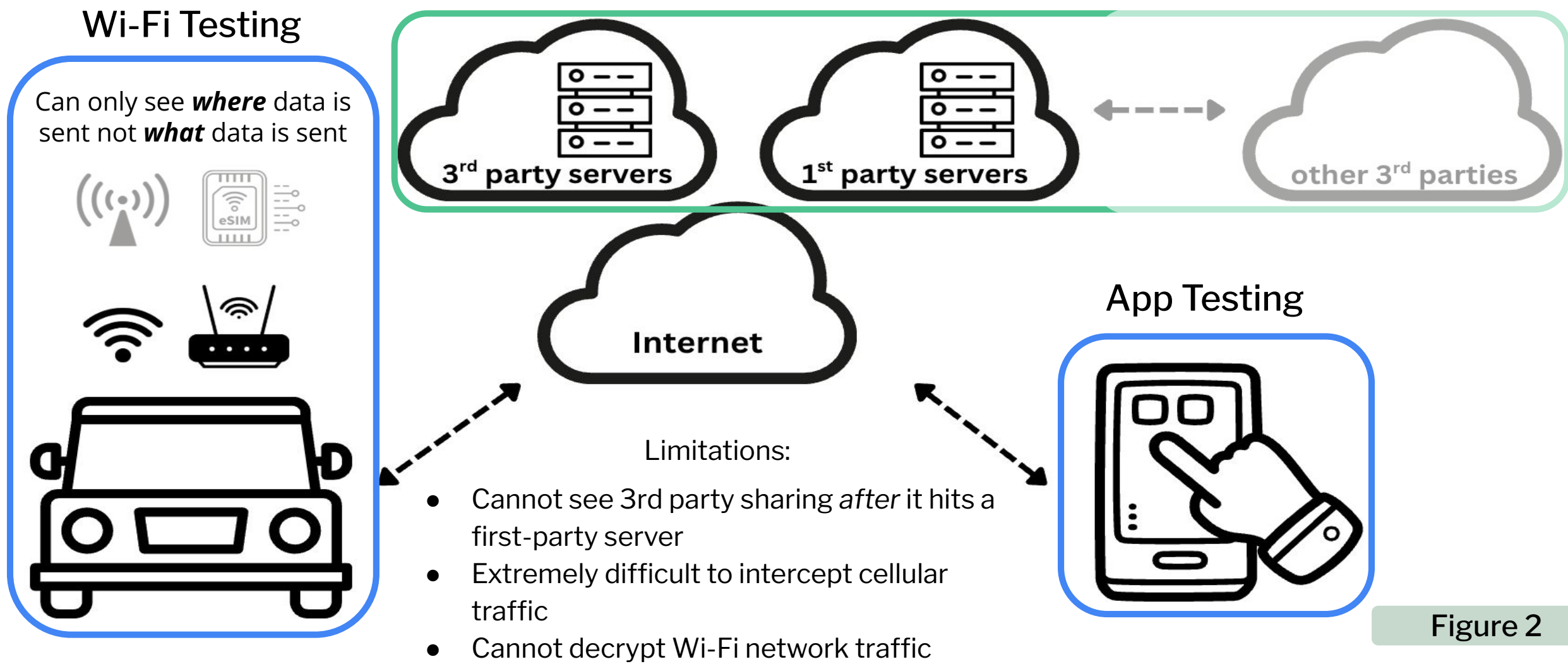


Figure 2

RESULTS

RQ1: PII and Sensitive Information

- Seven companion mobile apps shared PII with third-party trackers and advertisers. Bolded companies received VIN and at least one other form of PII from the same app.
- We observe that VINs are the most commonly shared PII within our sample, and that VINs were sent to major tech companies like Google, Microsoft, and Meta. We suspect that VINs are being used as stable, unique identifiers for vehicles, much like how IMEI or IDFA are used for the same purpose on smartphones.
- The combination of email and VIN may allow a major adtech firm like Adobe to link a person who owns a GM vehicle with their behavioral data and purchase history from other websites and mobile apps.

App Name	VIN	Location	Phone	Email
HondaLink	Amplitude	Amplitude		
Lincoln	ContentSquare			
MyNISSAN	Alchemer			Alchemer
MyCadillac	Adobe , Axciom ContentSquare , FullStory, Google, Meta, Microsoft, Pinterest, Snap, Yahoo	ContentSquare		Adobe , ContentSquare
MyChevrolet	Adobe , Axciom ContentSquare , FullStory, Google, Meta, Microsoft, Pinterest, Snap, Yahoo	Adobe		Adobe
myBuick	Adobe , Axciom ContentSquare , FullStory, Google, Meta, Microsoft, Pinterest, Snap, Yahoo			Adobe , ContentSquare
MyGMC	Adobe , Axciom ContentSquare , FullStory, Google, Meta, Microsoft, Pinterest, Snap, Yahoo	ContentSquare	ContentSquare	Adobe , ContentSquare

Table 1

RQ2/RQ3: Data Recipients

Table 2	Unique SLDs			
	Company Name	Cars	ATA	Integrated
	Alphabet Inc	13	11	12
	Amazon	12	3	7
	Spotify	8	3	4
	Liberty Media Corporation	7	0	1
	Cloudflare, Inc	6	0	1
	Microsoft	6	2	2
	TomTom International B.V.	5	0	1
	HERE Global B.V.	4	0	2
	Mapbox	4	0	1
	Meta Platforms, Inc.	3	2	0

App	Country	SLD	3P
Chrysler	Germany	xtn-lab.it	✓
Maserati Connect	Ireland	fcagcv.com	
MINI	Germany	bmwgroup.com	
My BMW	Germany	dynatrace.com	✓
My BMW	Germany	bmwgroup.com	
MyHyundai	Canada	streamtheworld.com	✓
MyHyundai	Netherlands	loopme.me	✓
Volvo Cars	Ireland	volvo.care	
Volvo Cars	Ireland	volvocars.com	

Table 3

The top 10 third-party companies contacted by vehicles in our sample over Wi-Fi, sorted by count of unique vehicles. We also present the total and unique number of domains associated with each company (Table 2.)

Table 4	Unique SLDs		
	App Name	ATA	Integrated
	MyCadillac	51	12
	myBuick	47	11
	My BMW	47	7
	Toyota	47	4
	myGMC	45	11
	...	...	...
	Land Rover	3	3
	Subaru Solterra Connect	3	2
	MercedesBenz	2	3
	MyHyundai	2	3
	GenesisIA	0	1

We see more tracking when testing companion apps compared to testing Wi-Fi (Table 4, 5.)

Table 5	Unique SLDs			
	Car	ATA	Integrated	Support
	Tesla Model 3	34	36	7
	Tesla Cybertruck	23	37	4
	Cadillac Lyriq	10	21	2
	Lucid Air	9	15	2
	Chevrolet Blazer	7	18	2
	...	...	...	...
	Nissan Ariya	0	1	1
	Toyota Corolla Cross	1	1	0
	Land Rover Range Rover	0	0	1
	Mercedes EQS	0	0	0
	Buick Envista	0	0	1

Six companion apps in our sample contacted a total of eight IP addresses outside the US (Table 3.)

CONCLUSIONS

- We tested 22 recently produced mass-market cars and 30 companion apps. Nearly all contact ad companies and trackers. It appears that U.S. consumers do not have a choice to purchase a new car that does *not* share their data with third parties and trackers by default.
- RQ1:** VINs may be used as permanent identifiers for cross-platform tracking.
- RQ2:** Cars contact ad companies and trackers just like our phones and the web.
- RQ3:** Rare for data to be transmitted internationally at the first hop.



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